

Modeling and Simulating the NEPSE Index Using Geometric Brownian Motion



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INSTITUTE OF SCIENCE AND TECHNOLOGY
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BY

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RECOMMENDATION

This is to recommend that **Aditya Kumar Sony**, (Symbol No. 500330008, T.U. Registration No. 5-2-33-137-2021), has carried out project work entitled “**Modeling and Simulating the NEPSE Index Using Geometric Brownian Motion**” for the requirement to the project work in Bachelor of Science (B.Sc.) degree in Mathematics under our supervision in the Department of Mathematics, Amrit Campus, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

To our knowledge, this work has not been submitted for any other degree.

He has fulfilled all the requirements laid down by the Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal for the submission of the project work for the partial fulfillment of Bachelor of Science (B.Sc.) degree.

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DECLARATION

This project work entitled “**Modeling and Simulating the NEPSE Index Using Geometric Brownian Motion**” is being submitted to the Department of Mathematics, Amrit Campus, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal for the partial fulfillment of the requirement to the project work in Bachelor of Science (B.Sc.) degree in Mathematics. This project work is carried out by me under the supervision of Asst. Prof. Dr. Khagendra Adhikari and co-supervision of Assoc. Prof. Dr. Surendra Raj Pathak in the Department of Mathematics, Amrit Campus, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

This work is original and has not been submitted earlier in part or full in this or any other form to any university or institute, here or elsewhere, for the award of any degree.

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LETTER OF FORWARD

[Date: 31/AUGUST/2026]

On the recommendation of **Asst. Prof. Dr. Khagendra Adhikari** and **Assoc. Prof. Dr. Surendra Raj Pathak**, this project work is submitted by Aditya Kumar Sony, Symbol No. 500330008, T.U. Registration No. 5-2-33-137-2021, entitled “**Modeling and Simulating the NEPSE Index Using Geometric Brownian Motion**” is forwarded by the Department of Mathematics, Amrit Campus, for the approval to the Evaluation Committee, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal.

He has fulfilled all the requirements laid down by the Institute of Science and Technology (IoST), Tribhuvan University (T.U.), Nepal for the project work.

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BOARD OF EXAMINATION AND CERTIFICATE OF APPROVAL

This project work (PRO-406) entitled “**Modeling and Simulating the NEPSE Index Using Geometric Brownian Motion**” by Aditya Kumar Sony (Symbol No. 500330008 and T.U. Registration No. 5-2-33-137-2021) under the supervision of Asst. Prof. Dr. Khagendra Adhikari and co-supervision of Assoc. Prof. Dr. Surendra Raj Pathak in the Department of Mathematics, Amrit Campus, Institute of Science and Technology (IoST), Tribhuvan University (T.U.), is hereby submitted for the partial fulfillment of the Bachelor of Science (B.Sc.) degree in Mathematics. This report has been accepted and forwarded to the Controller of Examination, Institute of Science and Technology, Tribhuvan University, Nepal for legal procedures.

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ABSTRACT

The stock market is uncertain and influenced by many economic and social factors, making accurate prediction difficult. The main objective of this project is to study NEPSE Index behavior using a stochastic model known as Geometric Brownian Motion (GBM).

In this project, historical data of the Nepal Stock Exchange (NEPSE) index is analyzed to estimate the drift and volatility parameters of the GBM model. Using these parameters, stock price paths are simulated and compared with actual market movements.

The results show that Geometric Brownian Motion is able to capture the random and continuous nature of stock price changes. Although it cannot predict exact future prices, the model provides useful insight into market behavior and risk. This study highlights the importance of stochastic models in understanding stock market dynamics.

LIST OF ACRONYMS AND ABBREVIATIONS

GBM	Geometric Brownian Motion
NEPSE	Nepal Stock Exchange
MAPE	Mean Absolute Percentage Error
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error

LIST OF SYMBOLS

S_t	Stock price or index value at time t
S_0	Initial stock price or index value
t	Time (in years)
μ	Drift parameter (expected return)
σ	Volatility of returns
W_t	Standard Brownian motion
dW_t	Increment of Brownian motion
Δt	Time step size
r_t	Log return at time t
$\ln$	Natural logarithm
$\mathbb{E}[\cdot]$	Expected value operator
$\mathbb{V}[\cdot]$	Variance operator
$\mathcal{N}(\mu, \sigma^2)$	Normal distribution with mean μ and variance σ^2
$\mathcal{N}(0, 1)$	Standard normal distribution
Z	Standard normal random variable, $Z \sim \mathcal{N}(0, 1)$

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Chapter 1

INTRODUCTION

The stock market is a place physical or online where people trade pieces of companies called stocks. When a company wants money to grow, it sells these shares to the public, and investors buy and sell them depending on market conditions and expectations. According to market demand share price rise and fall termed as Bull and Bear repectively.

1.1 Background

The modern stock market has evolved over centuries as a platform for raising capital and sharing investment risk. Its earliest form emerged in 17th-century Amsterdam, where the Dutch East India Company issued the first publicly tradable shares in 1602, giving rise to the Amsterdam Stock Exchange. In the late 18th century, the New York Stock Exchange (NYSE) was formed following the Buttonwood Agreement of 1792, and the London Stock Exchange was formally established in 1801, evolving from informal trading in coffee houses. Over time, stock markets worldwide have adapted to technological advancements, economic crises, and regulatory reforms, from the Great Depression in 1929 to the COVID-19 pandemic in 2020, shaping modern financial markets into complex, global systems that balance investment opportunities with the need for transparency and stability [20].

In Nepal, shares were first issued in 1936, but organized trading began only with the establishment of the Nepal Stock Exchange (NEPSE) in 1993. Since then, NEPSE has grown into the primary platform for securities trading in Nepal, reflecting the development of the domestic economy and the maturation of the country's financial sector. Today, NEPSE functions as a critical institution for capital mobilization and investment, providing insights into economic activity and

investor sentiment [21].

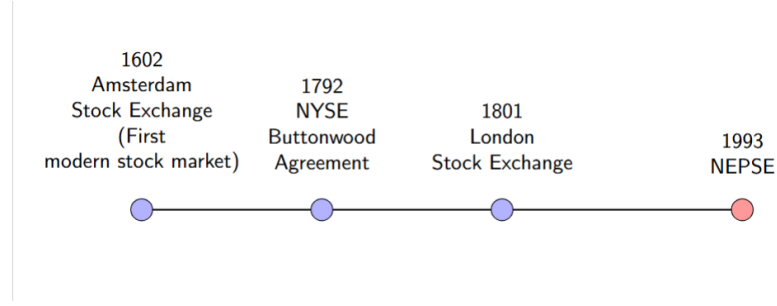


Figure 1.1: Origin of the Stock Market

1.2 NEPSE Index

The Nepal Stock Exchange (NEPSE) Index is the principal indicator of overall market performance in Nepal. Introduced to measure changes in the market value of listed companies, it provides investors, researchers, and policymakers with a single numerical representation of market trends.

The NEPSE Index is market capitalization-weighted, meaning that larger companies have a greater influence on its movements. The index's base year is 1994/95, with a base value of 100 points. It aggregates price movements from multiple sectors including banking, finance, insurance, hydropower, manufacturing, hotels, and trading. As such, fluctuations in the NEPSE Index reflect broader investor sentiment, economic expectations, and financial stability in Nepal [21, 23].

Recent years have seen significant volatility in the NEPSE Index due to political developments, policy changes, interest rate fluctuations, liquidity constraints, and global economic factors. These characteristics make the NEPSE Index particularly suitable for stochastic modeling and time-series analysis.

In this study, daily NEPSE Index closing values from the year 2020 to 2025 are transformed into logarithmic returns to meet the assumptions of continuous-time financial models. The Geometric Brownian Motion (GBM) model is then applied to simulate index behavior and estimate key parameters such as drift and volatility.

1.3 Problem Statement

The stock market is inherently uncertain, making it difficult for investors and financial analysts to predict future price movements accurately. The NEPSE Index,

which reflects the overall performance of the Nepalese stock market, experiences continuous fluctuations due to economic, political, and market-related factors. As a result, understanding its long-term behavior and quantifying the associated uncertainty remain significant challenges.

Traditional deterministic Simulating methods often fail to account for the random nature of stock price movements. Therefore, there is a need for a stochastic model that can describe the probabilistic behavior of the NEPSE Index and simulate its possible future paths. Geometric Brownian Motion (GBM) is a widely used stochastic process for modeling financial asset prices because it incorporates both the expected growth (drift) and market uncertainty (volatility).

This study addresses the problem of modeling the long-term behavior of the NEPSE Index using the GBM model. Specifically, it estimates the drift and volatility parameters from historical daily log-returns, simulates future index paths through Monte Carlo simulation, and evaluates the model's Simulating performance using out-of-sample validation. The study aims to determine whether the GBM model provides a reasonable representation of the long-term dynamics of the NEPSE Index despite its limitations in capturing sudden market shocks and time-varying volatility.

1.4 Objectives of the Study

The main objective of this study is to model the behavior of the NEPSE Index using the Geometric Brownian Motion (GBM) model.

The specific objectives are:

1. To estimate the drift (μ) and volatility (σ) parameters of the NEPSE Index using historical daily log-returns.
2. To simulate future paths of the NEPSE Index using the Geometric Brownian Motion (GBM) model and Monte Carlo simulation.
3. To evaluate the simulating performance of the GBM model by comparing the simulated values with the actual NEPSE Index using MAPE.

Chapter 2

LITERATURE REVIEW

Stock market modeling and Simulating are important areas of financial mathematics because of their applications in investment analysis, risk management, and financial decision-making. Since stock prices are affected by various economic and market factors, they contain a considerable degree of uncertainty. Stochastic models provide a useful mathematical framework for representing this uncertainty. Among these models, **Geometric Brownian Motion (GBM)** is widely used because of its simple structure and suitability for simulating financial asset prices.

GBM represents an asset price as a continuous-time stochastic process with a drift component and a random diffusion component. It is expressed as

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad (2.1)$$

where S_t is the asset price, μ is the drift, σ is the volatility, and W_t is a standard Wiener process. Under the assumptions of GBM, asset prices follow a log-normal distribution, which ensures that simulated prices remain positive [7]. The model became particularly important in financial mathematics through the Black–Scholes option pricing framework developed by Black and Scholes (1973) [1]. Later, Boyle (1977) demonstrated the use of Monte Carlo simulation in financial modeling, providing a practical approach for generating multiple possible outcomes of stochastic processes [3].

In practice, GBM parameters can be estimated from historical price data using logarithmic returns:

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right). \quad (2.2)$$

The estimated drift and volatility can then be used to generate future price paths through Monte Carlo simulation. This makes GBM useful for studying a range of

possible future market movements rather than producing only a single deterministic forecast.

Despite its simplicity, standard GBM has some limitations. It assumes constant drift and volatility and does not fully capture important characteristics of real financial markets, such as volatility clustering, heavy-tailed returns, and sudden price movements [4]. More advanced models, such as jump-diffusion and stochastic volatility models, have therefore been developed to address these limitations [2]. Nevertheless, GBM remains a useful benchmark because it is mathematically tractable, computationally efficient, and relatively easy to implement [5].

Recent studies have continued to apply GBM and Monte Carlo simulation to stock and stock-index Simulating. Sinha (2024) used GBM and Monte Carlo simulation to forecast major stock indices and demonstrated the practical use of simulated index paths for financial Simulating [16]. Tan and Ibrahim (2024) applied GBM with Monte Carlo simulation to stock-price prediction and evaluated the results using measures such as RMSE and MAPE [17].

GBM has also been applied to the Nepalese stock market. Thapa and Aryal (2021) used GBM to study the post-COVID-19 behavior of the NEPSE Index and demonstrated its usefulness for simulating possible future market scenarios [19]. Rai and Pantha also applied GBM to the simulation of future NEPSE Index values [18]. These studies provide a foundation for applying stochastic models to the Nepalese stock market.

Building on this existing work, the present study applies GBM to the NEPSE Index using recent historical data. Daily logarithmic returns are calculated from the index values, and the drift and volatility are estimated from the training data. The estimated parameters are then used to generate 1,000 Monte Carlo simulation paths. The simulated results are compared with the actual NEPSE Index during the validation period using MAPE. This provides an understanding of how well the standard GBM framework represents the observed behavior of the Nepalese stock market.

Chapter 3

MATHEMATICAL CONCEPTS

To model systems influenced by uncertainty, *stochastic differential equations* (SDEs) extend ordinary differential equations by incorporating random perturbations into the system dynamics. The mathematical foundation of SDEs relies on Brownian motion and stochastic calculus, particularly Itô's Lemma, which serves as the stochastic analogue of the classical chain rule [10, 13]. This chapter introduces these fundamental concepts and establishes the theoretical framework required for the Geometric Brownian Motion model developed in the subsequent chapter.

3.1 Ordinary and Stochastic Differential Equations

Ordinary differential equation can be used to form a deterministic model whereas in many real life phenomena, uncertainty cannot be ignored. For example, particles experience random molecular collisions, asset prices react unpredictably to information and investor behavior, and biological populations are influenced by random environmental factors. For modeling this, deterministic equations tend to underestimate variability and fail to capture observed fluctuations.

This limitation motivates the use of stochastic models, which explicitly account for randomness and provide a more realistic description of complex systems.

Ordinary Differential Equations(ODE)

A general first-order ODE is written as

$$\frac{dX(t)}{dt} = f(X(t), t),$$

where the rate of change of the state variable $X(t)$ is completely determined by its current value and time. Given an initial condition $X(0) = X_0$, the future trajectory of the system is uniquely determined.

Ordinary differential equations provide an appropriate mathematical framework for deterministic systems. However, they become inadequate when the evolution of the system is affected by random disturbances or unpredictable external factors [12].

Stochastic Differential Equations(SDE)

To model systems influenced by both systematic trends and random fluctuations, stochastic differential equations are employed. A general one-dimensional SDE is given by

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t,$$

where $\mu(X_t, t)$ represents the *drift term*, describing the average rate of change, and $\sigma(X_t, t)$ is the *diffusion coefficient*, measuring the intensity of randomness.

Interpretation

The drift term captures the long-term tendency of the process, while the diffusion term represents short-term random variations. The magnitude of the diffusion coefficient determines how strongly randomness affects the evolution of the system.

Fundamental Difference in ODE and SDE

In ordinary calculus, higher-order infinitesimal terms such as $(dt)^2$ are neglected. However, Brownian motion exhibits non-zero quadratic variation, given by

$$(dW_t)^2 = dt.$$

As a result, second-order terms contribute significantly to the dynamics. This fundamental difference invalidates the classical chain rule and necessitates a new calculus for stochastic processes.

3.2 Brownian Motion

The fundamental source of randomness in stochastic calculus is *Brownian motion*, also known as the Wiener process. It provides a mathematically rigorous model for

continuous random fluctuations and forms the backbone of stochastic differential equations [12].

Definition

A stochastic process $\{W_t\}_{t \geq 0}$ is called a standard Brownian motion if it satisfies the following properties:

1. $W_0 = 0$,
2. The increments $W_t - W_s$ are independent for $t > s$,
3. $W_t - W_s \sim \mathcal{N}(0, t - s)$,
4. The sample paths of W_t are continuous almost surely.

Although the paths of Brownian motion are continuous, they are nowhere differentiable. This irregularity is central to stochastic calculus and explains why classical calculus must be modified when randomness is present.

Integral Formulation of Stochastic Differential Equation

Since Brownian motion is not differentiable, SDEs are interpreted in integral form:

$$X_t = X_0 + \int_0^t \mu(X_s, s) ds + \int_0^t \sigma(X_s, s) dW_s.$$

The first integral is classical, whereas the second is an Itô stochastic integral.

3.3 Itô's Lemma

Assumptions

The function $f(x, t)$ is assumed to be once continuously differentiable in t and twice continuously differentiable in x , that is,

$$f \in C^{1,2}(\mathbb{R} \times [0, \infty)).$$

Statement of Itô's Lemma

If X_t satisfies

$$dX_t = \mu dt + \sigma dW_t,$$

then the process $Y_t = f(X_t, t)$ satisfies

$$dY_t = \left(f_t + \mu f_x + \frac{1}{2} \sigma^2 f_{xx} \right) dt + \sigma f_x dW_t.$$

The additional second-order term arises from the quadratic variation of Brownian motion and has no analogue in classical calculus. Itô's Lemma is a fundamental result in stochastic calculus used to derive the dynamics of stochastic processes [8, 10].

3.3.1 Proof of Itô's Lemma

To obtain the differential of $f(X_t, t)$, we consider the second-order Taylor expansion of f around (X_t, t) :

$$df = f_t dt + f_x dX_t + \frac{1}{2} f_{xx} (dX_t)^2.$$

Substituting the expression for dX_t into the above expansion yields

$$(dX_t)^2 = (\mu dt + \sigma dW_t)^2 = \mu^2 (dt)^2 + 2\mu\sigma dt dW_t + \sigma^2 (dW_t)^2.$$

Applying the fundamental multiplication rules of Itô calculus,

$$(dt)^2 = 0, \quad dt dW_t = 0, \quad (dW_t)^2 = dt,$$

the higher-order terms vanish, and we obtain

$$(dX_t)^2 = \sigma^2 dt.$$

Substituting this result back into the Taylor expansion, the differential of $f(X_t, t)$ becomes

$$df = \left(f_t + \mu f_x + \frac{1}{2} \sigma^2 f_{xx} \right) dt + \sigma f_x dW_t.$$

This expression constitutes Itô's Lemma for functions of a one-dimensional stochastic process with constant drift and volatility.

Example

Let

$$dX_t = \mu dt + \sigma dW_t, \quad f(X_t) = X_t^2.$$

Applying Itô's Lemma,

$$d(X_t^2) = (2\mu X_t + \sigma^2) dt + 2\sigma X_t dW_t.$$

This chapter introduced the mathematical foundations of stochastic modeling, including Brownian motion, stochastic differential equations, and Itô's Lemma. These tools provide a rigorous framework for analyzing systems influenced by uncertainty and form the theoretical basis for the stochastic models employed in subsequent chapters.

Chapter 4

GEOMETRIC BROWNIAN MOTION

Geometric Brownian Motion (GBM) is the most widely used stochastic process for modeling stock prices, market indices, and other financial assets [6, 7]. Its popularity stems from its strong theoretical foundation, mathematical tractability, and its ability to capture key empirical features of asset prices such as positivity and random proportional growth. This chapter develops the GBM model systematically, from its conceptual origins to its analytical solution and numerical simulation.

4.1 Definition of Geometric Brownian Motion

Let S_t denote the price of an asset or financial index at time t . The GBM model is defined by the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ represents the expected rate of return (drift), σ denotes the volatility, and W_t is standard Brownian motion. Both the deterministic growth and random fluctuations are proportional to the current price level [6, 10].

Why Geometric Brownian Motion?

When modeling asset prices, several empirical characteristics must be respected:

- Prices are strictly positive,
- Relative (percentage) changes are more meaningful than absolute changes,

- The magnitude of fluctuations increases with the price level.

Arithmetic Brownian motion fails to satisfy these properties, as it allows prices to become negative and treats changes additively. To overcome these limitations, asset prices are modeled using proportional changes, leading naturally to Geometric Brownian Motion [7].

Assumptions Underlying the GBM Model

The GBM framework relies on several simplifying assumptions:

- Asset returns are independent and identically distributed,
- Logarithmic returns follow a normal distribution,
- Drift and volatility remain constant over time,

While these assumptions facilitate analysis, they also introduce limitations, which are discussed later in this chapter.

4.2 Derivation of the GBM Solution Using Itô's Lemma

Consider the Geometric Brownian Motion (GBM) defined by the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (4.1)$$

where μ is the drift parameter, $\sigma > 0$ is the volatility parameter, and W_t is a standard Brownian motion. We assume $S_t > 0$ for all $t \geq 0$.

Logarithmic Transformation

To simplify the multiplicative structure of the SDE, we define the transformed process

$$Y_t = f(S_t) = \ln S_t.$$

The required derivatives of the function $f(S) = \ln S$ are

$$f'(S_t) = \frac{1}{S_t}, \quad f''(S_t) = -\frac{1}{S_t^2}.$$

Applying Itô's Lemma

For the process S_t satisfying the above SDE 4.1, Itô's Lemma states

$$df(S_t) = \left(f'(S_t)\mu S_t + \frac{1}{2}f''(S_t)\sigma^2 S_t^2 \right) dt + f'(S_t)\sigma S_t dW_t.$$

Substituting the derivatives of f into this expression yields

$$d(\ln S_t) = \left(\frac{1}{S_t}\mu S_t + \frac{1}{2}\left(-\frac{1}{S_t^2}\right)\sigma^2 S_t^2 \right) dt + \frac{1}{S_t}\sigma S_t dW_t.$$

Simplifying each term, we obtain

$$d(\ln S_t) = \left(\mu - \frac{1}{2}\sigma^2 \right) dt + \sigma dW_t.$$

Integrating both sides from time 0 to t gives

$$\ln S_t - \ln S_0 = \left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t.$$

Exponentiating both sides of the equation, we obtain the explicit solution of the Geometric Brownian Motion:

$$\boxed{S_t = S_0 \exp \left[\left(\mu - \frac{1}{2}\sigma^2 \right) t + \sigma W_t \right]} \quad (4.2)$$

This expression shows that S_t is log-normally distributed and remains strictly positive for all $t \geq 0$.

4.3 Discrete-Time Equation

A discrete-time equation is a mathematical model used for approximation and simulation of different future trajectories. It provides a step-by-step framework to analyze how a system evolves over time when changes occur at fixed intervals, making it especially useful for numerical simulations and Simulating.

Brownian Motion Increment

A standard Brownian motion (or Wiener process) $\{W_t\}_{t \geq 0}$ has independent increments satisfying

$$W_{t+\Delta t} - W_t \sim \mathcal{N}(0, \Delta t),$$

that is, the increment over a time interval of length Δt is normally distributed with mean 0 and variance Δt [6, 7].

Let

$$Z_t \sim \mathcal{N}(0, 1)$$

be a standard normal random variable. A basic property of the normal distribution states that if

$$Z \sim \mathcal{N}(0, 1),$$

then

$$\sigma Z \sim \mathcal{N}(0, \sigma^2).$$

Taking $\sigma = \sqrt{\Delta t}$ gives

$$\sqrt{\Delta t} Z_t \sim \mathcal{N}(0, \Delta t).$$

Since both $\Delta W_t = W_{t+\Delta t} - W_t$ and $\sqrt{\Delta t} Z_t$ follow the same normal distribution, the Brownian motion increment can be represented as

$$\boxed{\Delta W_t = \sqrt{\Delta t} Z_t, \quad Z_t \sim \mathcal{N}(0, 1),}$$

Using the closed-form solution 4.2, the discrete-time GBM update rule becomes

$$\boxed{S_{t+\Delta t} = S_t \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t \right]} \quad (4.3)$$

This expression forms the basis for numerical simulation of asset price paths.

4.4 Derivation of GBM Model Parameters

To estimate μ and σ we have, from section 4.2,

$$d \ln(S_t) = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t. \quad (4.4)$$

The term $\frac{1}{2} \sigma^2$ is the Itô correction term and arises from the second-order term in Itô's lemma. This result shows that the logarithm of a GBM process follows a Brownian motion with drift $\mu - \frac{1}{2} \sigma^2$ and volatility σ [10].

Integrating Equation (4.4) over a time interval Δt gives

$$\ln \left(\frac{S_{t+\Delta t}}{S_t} \right) = \left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma (W_{t+\Delta t} - W_t). \quad (4.5)$$

Since the increment of a Wiener process satisfies

$$W_{t+\Delta t} - W_t = \sqrt{\Delta t} Z_t, \quad Z_t \sim \mathcal{N}(0, 1), \quad (4.6)$$

Equation (4.5) can be written as

$$r_t = \ln \left(\frac{S_{t+\Delta t}}{S_t} \right) = \left(\mu - \frac{1}{2}\sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t. \quad (4.7)$$

Thus, the logarithmic return is normally distributed as

$$r_t \sim N \left[\left(\mu - \frac{1}{2}\sigma^2 \right) \Delta t, \sigma^2 \Delta t \right]. \quad (4.8)$$

This distribution provides the basis for estimating the GBM parameters from historical index data [24].

Derivation of Volatility Estimation Formula

Taking the variance of Equation (4.7), and using $Var(Z_t) = 1$, gives

$$Var(r_t) = \sigma^2 \Delta t. \quad (4.9)$$

Let s_r^2 denote the sample variance of the observed logarithmic returns. Therefore, the volatility parameter can be estimated as

$$\hat{\sigma} = \sqrt{\frac{s_r^2}{\Delta t}} = \frac{s_r}{\sqrt{\Delta t}} \quad (4.10)$$

where

$$s_r^2 = \frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2 \quad (4.11)$$

is the sample variance of the logarithmic returns and $\bar{r}$ is their sample mean.

For the daily NEPSE data used in this study, one trading year is assumed to contain 252 trading days. Hence,

$$\Delta t = \frac{1}{252}. \quad (4.12)$$

Consequently, the annualized volatility is obtained from the daily log-return standard deviation as

$$\hat{\sigma} = s_r \sqrt{252}. \quad (4.13)$$

Derivation of Drift Estimation Formula

From Equation (4.7), the expected value of the logarithmic return is

$$E[r_t] = \left(\mu - \frac{1}{2}\sigma^2 \right) \Delta t. \quad (4.14)$$

Replacing the theoretical expected log return by its sample mean $\bar{r}$ gives

$$\bar{r} = \left(\mu - \frac{1}{2}\sigma^2 \right) \Delta t. \quad (4.15)$$

Solving for the drift parameter μ gives

$$\boxed{\hat{\mu} = \frac{\bar{r}}{\Delta t} + \frac{1}{2}\hat{\sigma}^2} \quad (4.16)$$

where $\bar{r}$ is the mean of the observed daily logarithmic returns and $\hat{\sigma}$ is the estimated annualized volatility.

Therefore, the two GBM parameters used in the simulation are estimated from the historical logarithmic returns as

$$\boxed{\hat{\sigma} = \frac{s_r}{\sqrt{\Delta t}}} \quad \text{and} \quad \boxed{\hat{\mu} = \frac{\bar{r}}{\Delta t} + \frac{1}{2}\hat{\sigma}^2}. \quad (4.17)$$

4.5 Expected Value of the Index Under GBM

The exact solution of the GBM equation is given by

$$S_T = S_0 \exp \left[\left(\mu - \frac{\sigma^2}{2} \right) T + \sigma W_T \right]. \quad (4.18)$$

where μ is the drift coefficient, σ is the volatility, and W_t is a standard Brownian motion.

Taking the expectation on both sides gives

$$\mathbb{E}[S_T] = S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)T} \mathbb{E}[e^{\sigma W_T}]. \quad (4.19)$$

Since $W_T \sim N(0, T)$, the moment-generating function of a normal random variable gives

$$\mathbb{E}[e^{\sigma W_T}] = e^{\frac{1}{2}\sigma^2 T}. \quad (4.20)$$

Therefore,

$$\begin{aligned} \mathbb{E}[S_T] &= S_0 e^{\left(\mu - \frac{\sigma^2}{2}\right)T} e^{\frac{1}{2}\sigma^2 T} \\ &= S_0 e^{\mu T}. \end{aligned} \quad (4.21)$$

Hence, the expected value of the NEPSE Index after T years under the GBM model is

$$\boxed{\mathbb{E}[S_T] = S_0 e^{\mu T}}. \quad (4.22)$$

4.6 Monte Carlo Simulation from GBM Model

Monte Carlo simulation is widely used for approximating stochastic models in finance [3,5]. This Monte Carlo approach enables analysts to study the distribution of future stock prices rather than relying on a single prediction.

Monte Carlo simulation uses the discrete-time GBM formula:

$$S_{t+\Delta t} = S_t \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t \right],$$

where $Z_t \sim \mathcal{N}(0, 1)$ are independent standard normal random numbers.

Simulation steps:

1. Start with the initial price S_0 .
2. Generate a random number Z_t for each time step.
3. Update the stock price using the GBM formula.
4. Repeat to simulate a single path.
5. Repeat the process many times to create multiple possible paths.

4.6.1 Illustrative Example: Simulating Stock Prices

To illustrate the application of Geometric Brownian Motion (GBM), consider a stock whose price follows the stochastic differential equation

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where μ is the drift parameter, σ is the volatility, and W_t is a standard Brownian motion.

Simulation Parameters

Suppose the stock has the following characteristics:

Initial stock price	$S_0 = 100$
Expected annual return	$\mu = 0.10$
Annual volatility	$\sigma = 0.20$
Time horizon	$T = 1$ year

Using the closed-form solution of GBM,

$$S_T = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma W_T \right],$$

the stock price after one year depends on the realization of the Brownian motion W_T .

Expected Value

For the above parameters,

$$\mathbb{E}[S_T] = S_0 e^{\mu T} = 100 e^{0.10} \approx 110.52.$$

Thus, while the average stock price after one year is expected to be approximately Rs.110.52, the actual price may vary considerably due to random market fluctuations.

Calculating Single Simulated Price

Suppose one realization of the Brownian motion is

$$W_T = 0.65.$$

Substituting this value into the GBM solution gives

$$\begin{aligned} S_T &= 100 \exp \left[\left(0.10 - \frac{1}{2} \times 0.20^2 \right) + 0.20(0.65) \right] \\ &= 100 \exp(0.08 + 0.13) \\ &= 100 e^{0.21} \\ &\approx 123.37. \end{aligned}$$

This represents one possible future stock price generated by the GBM model. Different realizations of W_T produce different future prices, reflecting the inherent uncertainty of financial markets.

Simulation

A single simulation provides only one possible future outcome. In practice, the GBM model is simulated repeatedly using independently generated standard normal random variables to obtain many possible price paths.

Figure 4.1 illustrates several simulated price paths generated from the GBM model. Each curve represents one possible evolution of the stock price over time. Although all paths originate from the same initial value, they diverge because of random Brownian shocks. The increasing dispersion of the paths reflects the cumulative effect of volatility, while their overall upward trend is governed by the positive drift parameter.

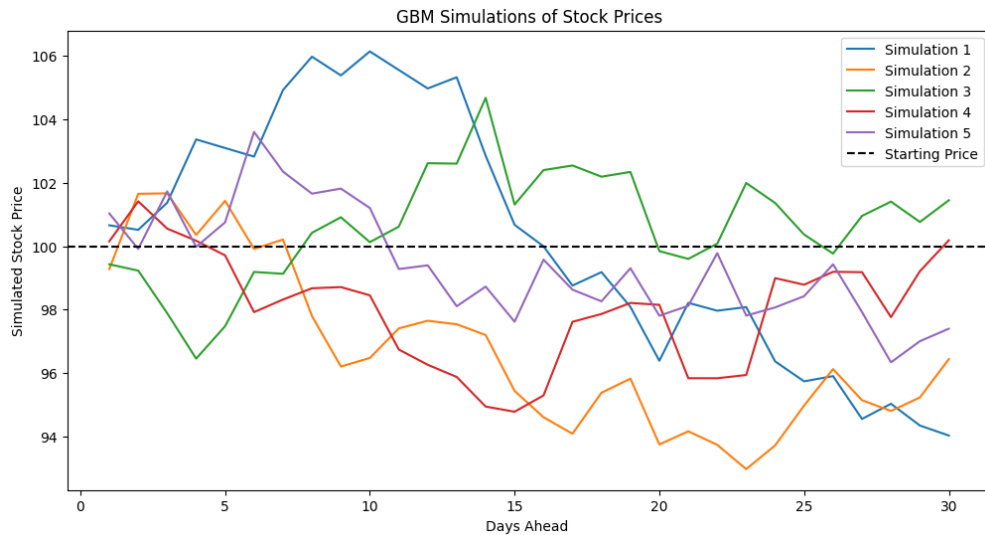


Figure 4.1: Monte Carlo simulation of multiple stock price paths using Geometric Brownian Motion.

4.7 Applications and Limitations

Geometric Brownian Motion is widely used in:

- Stock price and index simulation,
- Option pricing (Black–Scholes model),
- Risk management and scenario analysis.

However, GBM cannot capture:

- Volatility clustering,
- Sudden market jumps,
- Time-varying volatility.

Despite these limitations, GBM remains a fundamental benchmark model.

This chapter developed the mathematical formulation of Geometric Brownian Motion as a stochastic model for modeling the Nepse index behaviour.

Chapter 5

METHODOLOGY

5.1 Overview of the Methodology

This chapter describes the methodology used to model and forecast the Nepal Stock Exchange (NEPSE) Index using the Geometric Brownian Motion (GBM) model 4.3. The study consists of five major stages:

- Data collection,
- Data preprocessing: Computing log-return,
- Estimation of the Model parameters,
- Simulating future index paths,
- Model Validation

5.2 Data Description

The complete dataset contains trading-day observations only, excluding weekends and public holidays when the stock exchange remains closed.

The data were collected from multiple publicly available sources, including the official NEPSE website and the Kaggle dataset [22] .

Table 5.1: Summary of the NEPSE Dataset

Dataset	Period	Purpose
Training Data	01-01-2020 to 30-06-2025	Estimation of GBM parameters
Validation Data	01-07-2025 to 30-06-2026	Model validation and Simulating accuracy

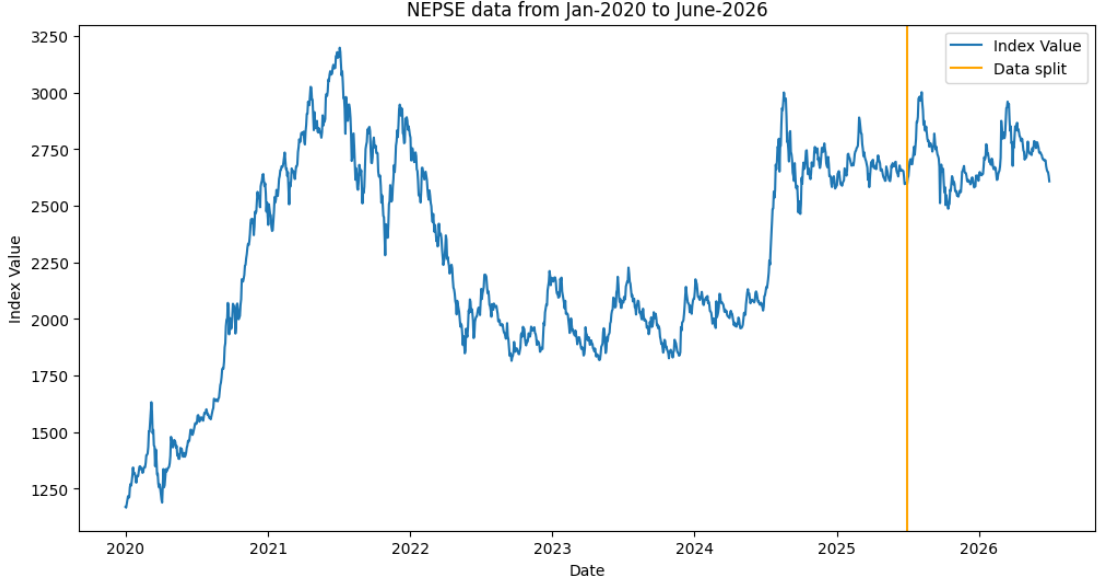


Figure 5.1: NEPSE Index Data

5.3 Computing Log>Returns from the Data

Let S_t denote the closing value of the NEPSE Index on trading day t . The daily logarithmic return, or log-return, is calculated as

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right), \quad (5.1)$$

where $\ln(\cdot)$ denotes the natural logarithm and S_{t-1} represents the closing index value on the previous trading day. Log-returns are used because they are additive over time and are consistent with the lognormal framework assumed by the Geometric Brownian Motion (GBM) model.

Before calculating the log-returns, the historical NEPSE Index data were pre-processed to ensure consistency and chronological ordering. The following steps were performed:

1. The date variable was converted into a standard datetime format.
2. The observations were sorted chronologically according to the trading date.
3. Missing values in the relevant variables were identified and checked.
4. Daily logarithmic returns were calculated using Equation (5.1).
5. The first observation was excluded from the return series because a previous trading-day index value is not available for calculating its log-return.

After preprocessing, the resulting log-return series was used for the estimation of the parameters required for the GBM model.

Sample of Computed Log>Returns

Table 5.2 presents a sample of the NEPSE Index observations together with their corresponding daily log-returns.

Table 5.2: Sample of NEPSE Index Data with Log>Returns

Date	Index Value	Log-Return (r_t)
02-01-2020	1166.21	-0.002817
05-01-2020	1182.31	0.013711
06-01-2020	1200.69	0.015426
07-01-2020	1216.92	0.013427
08-01-2020	1209.60	-0.006033
$\vdots$	$\vdots$	$\vdots$
30-06-2025	2631.48	0.0035022

Log-Return Series of the Training Data

The calculated daily log-returns of the training dataset are shown in Figure 5.2. The plot illustrates the day-to-day fluctuations in the NEPSE Index and provides an initial view of the variability of market returns over the study period. These returns are subsequently used to estimate the drift and volatility parameters of the GBM model.

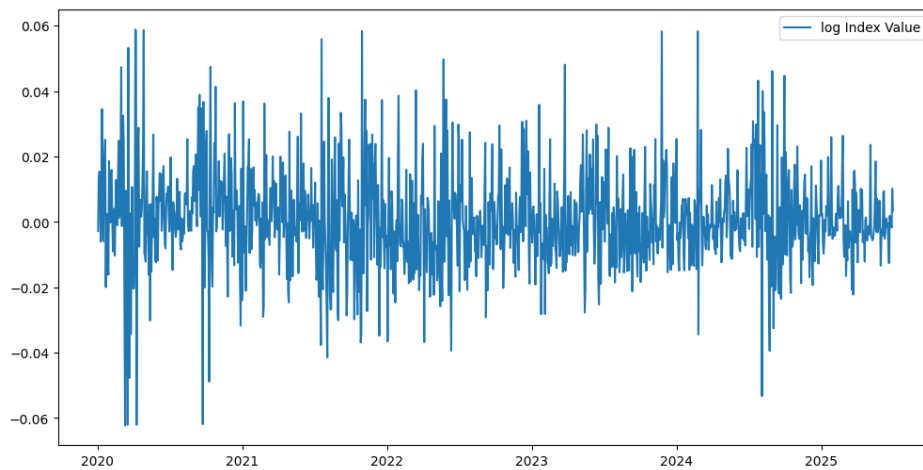


Figure 5.2: Daily log-returns of the NEPSE Index training data.

5.4 Estimation of Model Parameters

There are two parameters which we estimate from our training data, they are volatility(σ) and drift (μ) as shown in the model 4.3 for the Simulating of Nepse Index paths.

Volatility Estimation

Volatility is estimated as the sample standard deviation of the daily log-returns:

$$\sigma_d = \sqrt{\frac{1}{N-1} \sum_{t=1}^N (r_t - \bar{r})^2}, \quad (5.2)$$

where $\bar{r}$ is the sample mean of the daily log-returns and N is the number of return observations.

For the training dataset, the estimated daily volatility is

$$\sigma_d \approx 0.01515.$$

Since the analysis considers 223 trading days as one trading year, the daily volatility is annualized as

$$\sigma = \sigma_d \sqrt{223}. \quad (5.3)$$

Therefore,

$$\sigma = 0.01515 \sqrt{223} \approx 0.2263.$$

Thus, the estimated annualized volatility is approximately

$$\boxed{\sigma \approx 22.63\%}.$$

Drift Estimation

The drift parameter is estimated using

$$\mu = \frac{\bar{r}}{\Delta t} + \frac{1}{2} \sigma^2, \quad (5.4)$$

where $\Delta t = 1/223$ represents one trading day as a fraction of a year. This results in

$$\mu \approx 0.1715.$$

5.5 Index Simulating Algorithm

The Monte Carlo simulation method is employed to forecast 1,000 possible future paths of the NEPSE Index based on the Geometric Brownian Motion (GBM) model described in Section 4.6. The overall Simulating procedure, from the preparation of historical data to the generation of simulated index paths, is summarized in the flowchart shown in Figure 5.3.

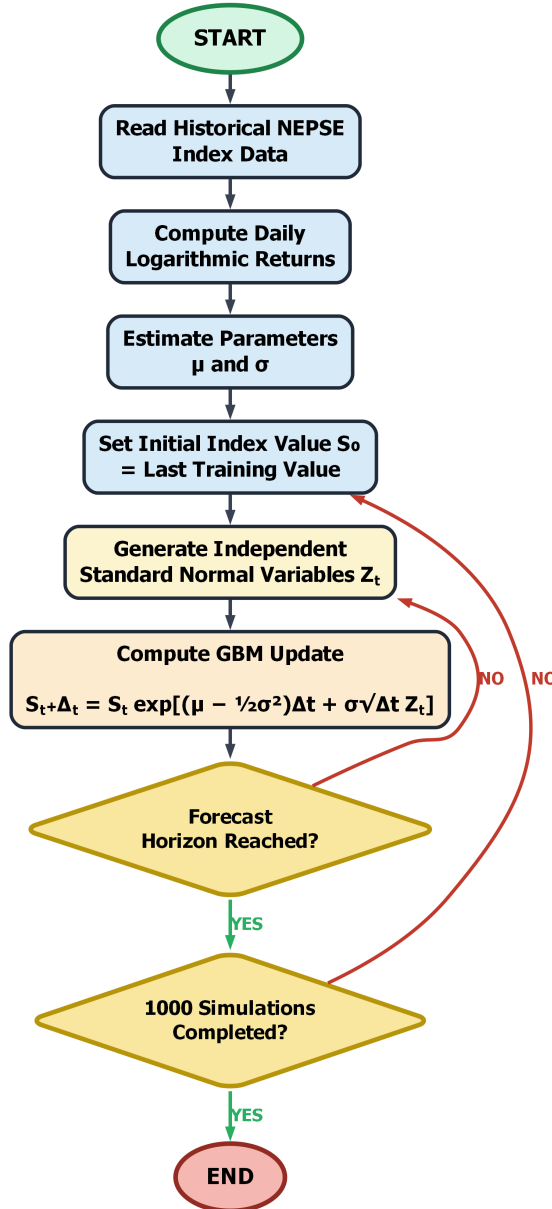


Figure 5.3: Flowchart of the Monte Carlo-GBM index Simulating algorithm.

5.6 Model Evaluation

The Simulating performance of the GBM model was assessed by comparing the average simulated index values with the actual observed NEPSE Index during the validation period.

The average simulated trajectory was obtained by taking the arithmetic mean of the simulated values across all Monte Carlo paths at each trading day.

Mean Absolute Percentage Error (MAPE)

The Mean Absolute Percentage Error (MAPE) is a commonly used metric for evaluating the accuracy of Simulating models [15]. It measures the average percentage difference between the actual and predicted values and is defined as

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{A_i - F_i}{A_i} \right|, \quad (5.5)$$

where n is the number of observations, A_i is the actual value, and F_i is the predicted value.

In this study, MAPE is used to evaluate the Simulating performance of the Geometric Brownian Motion (GBM) model by comparing the actual NEPSE Index values with the mean of the simulated GBM paths during the validation period. According to [14], a MAPE value below 10% indicates excellent Simulating accuracy, 10–20% indicates good accuracy, 20–50% indicates reasonable accuracy, and values above 50% indicate poor Simulating performance.

This chapter presented the methodology adopted for modeling the NEPSE Index using Geometric Brownian Motion. The results obtained from this methodology are presented in the following chapter.

Chapter 6

RESULTS AND DISCUSSION

6.1 Overview of Results

This chapter presents and interprets the results obtained by applying the Geometric Brownian Motion (GBM) model 4.3 to the Nepal Stock Exchange (NEPSE) index using daily data from 2020 to 2025-June and validate the simulated 1 year return with test data from 1-July-2025 to 30-June-2026.

6.2 Descriptive Statistics of NEPSE Returns

Table 6.1: Descriptive Statistics of NEPSE Daily Log>Returns (2020–2025)

	Index Value	Log-Return
Count	1240	1240
Mean	2230.6076	0.00065
Std. Dev.	457.7188	0.01515
Minimum	1166.21	-0.062262
25%	1939.3975	-0.007693
Median	2112.38	-0.000211
75%	2651.805	0.007476
Maximum	3198.6	0.058847

Table 6.1 presents the descriptive statistics of the NEPSE Index and its daily log-returns from 2020 to 2025 based on **1,240 observations**. The average NEPSE Index was **2230.61**, with values ranging from **1166.21** to **3198.60**, indicating noticeable fluctuations during the study period.

The mean daily log-return was **0.00065**, indicating a slight positive average daily growth. The standard deviation of **0.01515** suggests moderate daily mar-

ket volatility. The minimum and maximum daily log-returns were -0.062262 and **0.058847**, respectively, reflecting occasional large market movements. Since the median log-return (-0.000211) is slightly lower than the mean, the return distribution appears to be slightly positively skewed. Furthermore, the interquartile range indicates that 50% of the daily log-returns lie between -0.007693 and **0.007476**, suggesting that most daily price changes were relatively small.

Overall, the descriptive statistics indicate that the NEPSE Index experienced moderate volatility with a slight upward trend, making the data suitable for estimating the drift and volatility parameters required for the Geometric Brownian Motion (GBM) model.

6.3 Distribution of Log-Returns

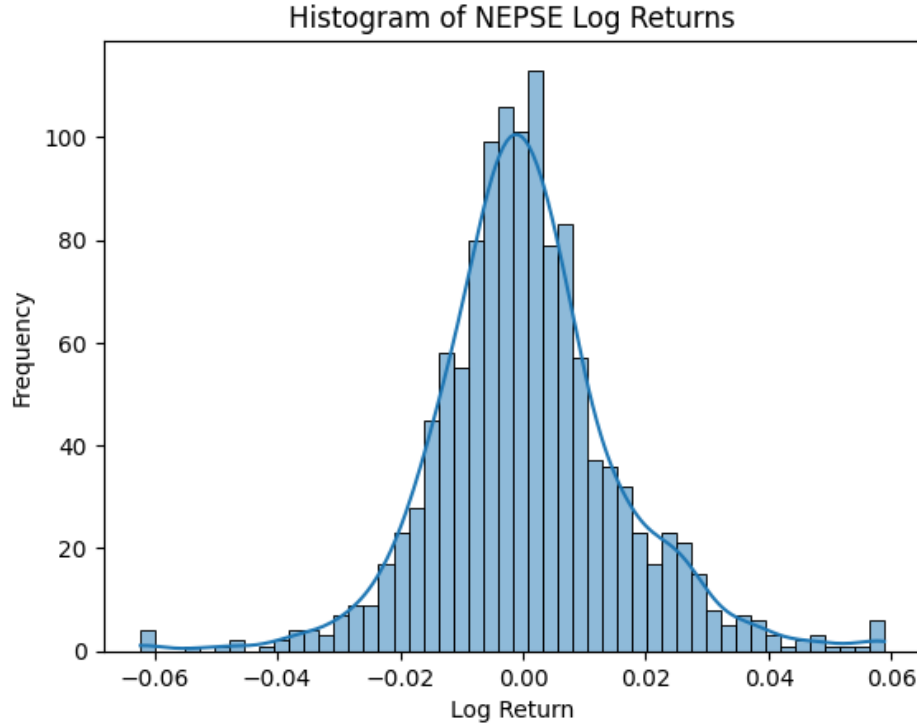


Figure 6.1: Distribution of NEPSE Log-Returns

Figure 6.1 illustrates the empirical distribution of NEPSE log-returns. The distribution is approximately bell-shaped, indicating a reasonable alignment with the normal distribution in the central region. This supports the use of GBM as a first-order approximation of average market behavior.

6.4 Estimated GBM Parameters

Table 6.2: Estimated GBM Parameters for the NEPSE Index

Parameter	Estimated Value
Drift (μ)	0.1715
Volatility (σ)	0.2263

Table 6.2 presents the estimated parameters of the Geometric Brownian Motion (GBM) model for the NEPSE Index. The estimated drift parameter (μ) is **0.1715**, indicating a positive long-term growth trend in the index. The estimated volatility (σ) is **0.2263**, representing the annual variability in the index returns. These estimated parameters serve as the key inputs for simulating future NEPSE Index paths using the GBM model.

6.5 Simulating of NEPSE Index Paths

Using the estimated parameters, 1000 NEPSE index paths were simulated through Monte Carlo methods. The initial index value was set equal to the observed NEPSE index level on 30-06-2025, namely $S_0 = 2631.48$ and the forecast horizon was same as number of validation data (223).

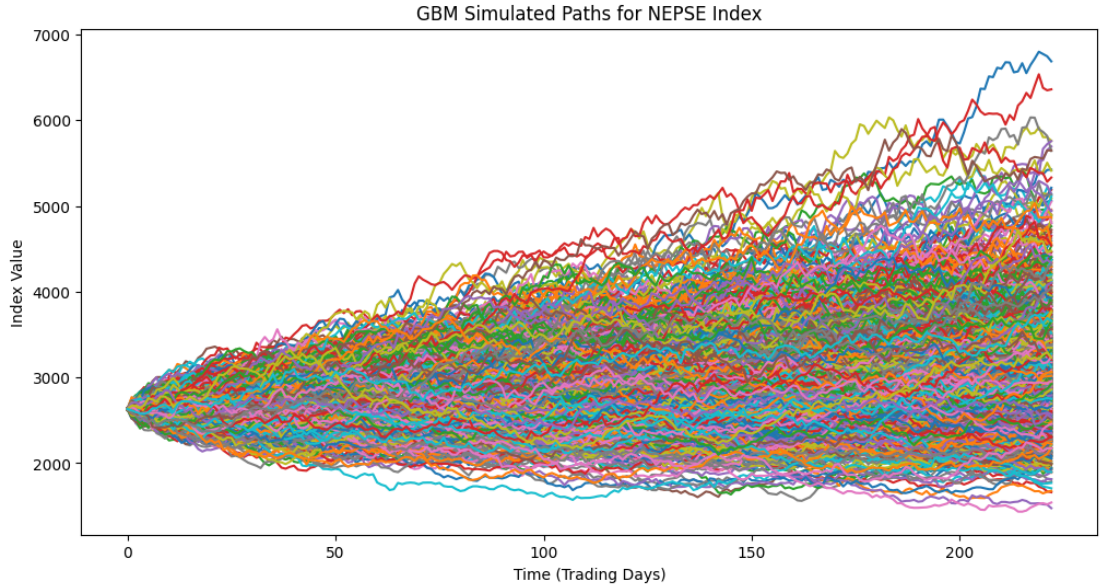


Figure 6.2: Simulated NEPSE Index Paths Using the GBM Model

The simulated trajectories illustrate the stochastic nature of the GBM process by showing the different possible paths. The increasing dispersion of simulated

paths over time demonstrates how uncertainty accumulates as the forecast horizon extends.

6.5.1 Expected Value After 223 Trading Days

For the validation period, the forecast horizon consists of 223 trading days which is one year in stock trading in Nepal.

From the GBM model, the theoretical expected value of the index at time T is given by

$$\mathbb{E}[S_T] = S_0 e^{\mu T}. \quad (6.1)$$

Substituting $T = 223 = 1$ year gives

$$\mathbb{E}[S_{223}] = S_0 e^{\mu \cdot 1}. \quad (6.2)$$

Using the estimated initial index value and drift parameter from this study,

$$S_0 = 2633.76, \quad \mu = 0.1715,$$

the expected value after 223 trading days is

$$\mathbb{E}[S_{223}] = 2633.76 e^{0.1715} \quad (6.3)$$

$$\approx 3126.5. \quad (6.4)$$

Therefore, the theoretical expected value of the NEPSE Index after 223 trading days under the GBM model is approximately

$$\boxed{\mathbb{E}[S_{223}] \approx 3126.5}. \quad (6.5)$$

This value represents the theoretical mean of the GBM distribution at the end of the 223-trading-day forecast horizon.

6.6 Model Evaluation Results

Actual Path and Mean of Simulated Paths

The GBM model was validated using **223 out-of-sample observations** from **1 July 2025** to **30 June 2026** as shown in Figure 6.3. Although the GBM model

does not capture all short-term market fluctuations due to its assumptions of constant drift and volatility, it provides a reasonable approximation of the overall market trend and uncertainty. Therefore, the results suggest that the GBM model is suitable for long-term stochastic modeling of the NEPSE Index, while its ability to predict sudden market movements remains limited.

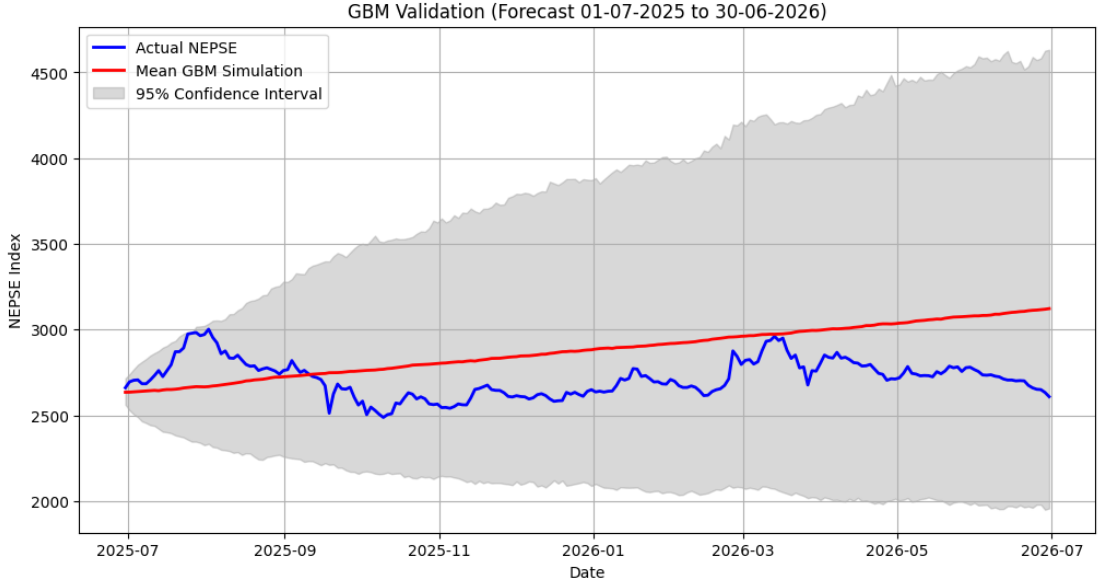


Figure 6.3: Observed NEPSE Index and Mean Simulated GBM Path

Mean Absolute Percentage Error (MAPE)

The Mean Absolute Percentage Error (MAPE) is used to evaluate the Simulating accuracy of the GBM model. It is calculated as:

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{A_i - F_i}{A_i} \right|, \quad (6.6)$$

where $n = 223$ is the number of validation observations, A_i is the actual NEPSE Index value, and F_i is the corresponding predicted value for the i^{th} observation ($i = 1, 2, \dots, n$).

The computed MAPE for the validation period is:

$$\text{MAPE} = 7.94\%.$$

Also,

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |A_i - F_i|, \quad (6.7)$$

The computed MAE for the validation period is:

$$\text{MAE} = 214.11.$$

And,

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (A_i - F_i)^2}, \quad (6.8)$$

The computed RMSE for the validation period is:

$$\text{RMSE} = 237.54.$$

The model achieved a **Mean Absolute Percentage Error (MAPE)** of **7.94%**. The low MAPE indicates that the simulated values differ from the actual NEPSE Index by only **7.94%** on average, demonstrating good Simulating accuracy.

This chapter analyzed and interpreted the empirical results of modeling the NEPSE index using the Geometric Brownian Motion framework.

Chapter 7

CONCLUSIONS AND RECOMMENDATION

7.1 Conclusion

This study applied the Geometric Brownian Motion (GBM) model to simulate and analyze the future movement of the NEPSE Index using daily closing prices from 2020 to 2025 June. The drift and volatility parameters were estimated from daily logarithmic returns and used to generate multiple simulated index paths through Monte Carlo simulation.

The descriptive analysis showed that the NEPSE Index exhibited moderate daily volatility with a slight positive average return over the study period. The estimated GBM parameters were a drift of $\mu = 0.1715$ and a volatility of $\sigma = 0.2263$, indicating a positive long-term growth trend with moderate market fluctuations.

To evaluate the performance of the model, out-of-sample validation was carried out using 223 observations from 1 July 2025 to 30 June 2026. The model achieved a Mean Absolute Percentage Error (MAPE) of 7.94%. The low MAPE value indicates that the simulated values closely matched the actual NEPSE Index values, demonstrating good Simulating performance. These results suggest that the GBM model is effective in capturing the overall market trend and uncertainty over longer time horizons.

7.1.1 Discussion of Model Limitations

Several limitations arise when applying the GBM model to the NEPSE index:

- Only the data from 2020 to 2026 June is used,

- Volatility clustering is not captured,
- Sudden market fluctuations caused by economic or political events are ignored, and
- The assumption of constant parameters may not hold over long time horizons.

These limitations are particularly important in the context of emerging markets such as Nepal.

7.2 Recommendations

Based on the findings of this study, the following recommendations are proposed:

1. Geometric Brownian Motion (GBM) can be used as a simple and effective stochastic model for understanding the long-term behavior of the NEPSE Index. However, it should be applied with caution for short-term Simulating due to its assumption of constant drift and volatility.
2. Future studies should consider advanced stochastic models, such as jump-diffusion models, stochastic volatility models, or GARCH models, to better capture sudden market movements and time-varying volatility.
3. The Simulating performance of GBM can be further improved by incorporating macroeconomic variables, interest rates, inflation, and other market indicators that influence stock price movements.
4. Future research may use a longer historical dataset and validate the model over multiple out-of-sample periods to obtain more robust and reliable Simulating results.
5. The proposed methodology can also be extended to model and forecast individual stocks or other financial indices listed on the Nepal Stock Exchange.

In conclusion, while GBM does not capture all the complexities of the NEPSE market, it provides a clear, practical framework for analyzing trends and serves as a stepping stone for more detailed and realistic financial models in the future.

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Appendix A

Appendix: Python Code Used in the Project

Listing A.1: Sample Python Code used in this Project for GBM

```
1  #importing required libraries
2
3  import numpy as np
4  import pandas as pd
5  import seaborn as sns
6  import matplotlib.pyplot as plt
7  import scipy.stats as stats
8  from sklearn.metrics import mean_absolute_percentage_error
9
10 # Loading NEPSE data here
11 df=pd.read_excel(r'C:\Users\legen\Downloads\2020-2026-NEPSE.
    xlsx')
12 df.head(10)
13
14 # Data Visualization
15 dates = pd.date_range(
16     start='2020-01-01',
17     periods=num_days,
18     end='30-06-2026'
19 )
20 plt.figure(figsize=(15,5))
21 plt.plot(dates,df['Index Value'],label='Index Value')
22 plt.axvline(pd.Timestamp('2025-07'),color='orange',label='
    Data split')
```

```

23 plt.xlabel('Date')
24 plt.ylabel('Index Value')
25 plt.title('NEPSE data from Jan-2020 to June-2026')
26 plt.legend()
27 plt.show()
28
29 # Split into train and validation
30 # -----
31 train = df[df['Date'] <= '2025-06-30']
32 test = df[df['Date'] > '2025-06-30']
33
34 print("Training observations :", len(train))
35 print("Validation observations :", len(test))
36
37 # Log Return
38 train['log_return']=np.log(train['Index Value']/train['Index
    Value'].shift(1))
39
40 #log return visual
41 date1=pd.date_range(start='01-01-2020',
42                     end='30-06-2025',
43                     periods=len(train))
44
45 plt.figure(figsize=(15,5))
46 plt.plot(date1,train['log_return'],label='log Index Value')
47 plt.legend()
48 plt.show()
49
50 #checking distribution of Log return
51 #for histogram
52 plt.hist(train['log_return'], bins=50)
53 plt.title("Histogram of NEPSE Log Returns")
54 plt.xlabel("Log Return")
55 plt.ylabel("Frequency")
56 plt.show()
57
58 #Descriptive Statistics of data
59 train.describe()
60
61 T = 252          # number of days to predict

```

```

62 dt = 1 / 252    # daily step
63 n_simulations = 1000
64
65 # Parameters
66 r_m = train['log_return'].mean()    # mean return
67 sigma = train['log_return'].std()    # volatility
68 mu=r_m/dt + 0.5*(sigma**2)
69
70 # Create simulation matrix
71 simulations = np.zeros((T, n_simulations))
72
73 # Generate GBM paths
74 for j in range(n_simulations):
75
76     prices = [S0]
77
78     for i in range(1, T):
79
80         Z = np.random.normal(0, 1)
81
82         next_price = prices[-1] * np.exp(
83             (mu - 0.5 * sigma**2) * dt +
84             sigma * np.sqrt(dt) * Z
85         )
86
87         prices.append(next_price)
88
89     simulations[:, j] = prices
90
91 # Plot simulations
92 plt.figure(figsize=(10, 6))
93
94 for j in range(n_simulations):
95     plt.plot(simulations[:, j])
96
97 plt.title("GBM Simulated Paths for NEPSE Index")
98 plt.xlabel("Time (Trading Days)")
99 plt.ylabel("Index Value")
100 plt.show()
101

```

```

102 #Out of Sample Validation
103
104 # Initial price
105
106 S0 = train['Index Value'].iloc[-1]
107
108 # Number of validation days
109 N = len(test)
110
111 # Number of simulations
112 n_sim = 1000
113
114 np.random.seed(42)
115
116 # Monte Carlo Simulation
117
118 paths = np.zeros((N+1, n_sim))
119 paths[0] = S0
120
121 for j in range(n_sim):
122     for i in range(1, N+1):
123         Z = np.random.normal()
124         paths[i, j] = paths[i-1, j] * np.exp(
125             (mu - 0.5*sigma**2)*dt +
126             sigma*np.sqrt(dt)*Z
127         )
128
129 # Remove initial value
130 paths = paths[1:]
131
132 # Mean prediction
133
134 mean_path = paths.mean(axis=1)
135
136 lower = np.percentile(paths, 2.5, axis=1)
137 upper = np.percentile(paths, 97.5, axis=1)
138
139 actual = test['Index Value'].values
140
141 # Validation Metrics

```

```

142
143 mape=mean_absolute_percentage_error(actual,mean_path)*100
144 print(f'MAPE : {mape:.2f}%')
145 # -----
146 # Plot
147 # -----
148 plt.figure(figsize=(12,6))
149
150 date2 = pd.date_range(
151     start='30-06-2025',
152     periods=len(test),
153     end='30-06-2026'
154 )
155 plt.plot(date2, actual,
156     color='blue',
157     linewidth=2,
158     label='Actual NEPSE')
159
160 plt.plot(date2, mean_path,
161     color='red',
162     linewidth=2,
163     label='Mean GBM Simulation')
164
165 plt.fill_between(date2,
166     lower,
167     upper,
168     color='gray',
169     alpha=0.3,
170     label='95% Confidence Interval')
171
172 plt.xlabel("Date")
173 plt.ylabel("NEPSE Index")
174 plt.title("GBM Validation (Forecast 01-07-2025 to
175     30-06-2026)")
175 plt.legend()
176 plt.grid(True)
177
178 plt.show()

```